# Tutorial 7 - Digital Systems

1. Consider the circuit shown in Figure 1. We want this circuit to count down from 15 to 0. Suggest what changes are required and draw the new circuit. Also, explain the operation of your (new) circuit.

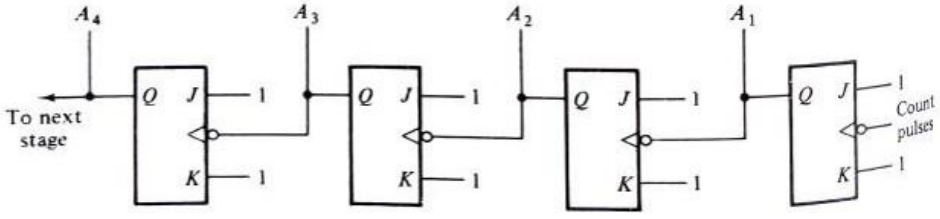



Figure 1: Ripple Counter

2. An Up-Down counter circuit is shown in Figure 2. Study the behaviour of this circuit for all possible inputs (binary values of Up and Down).

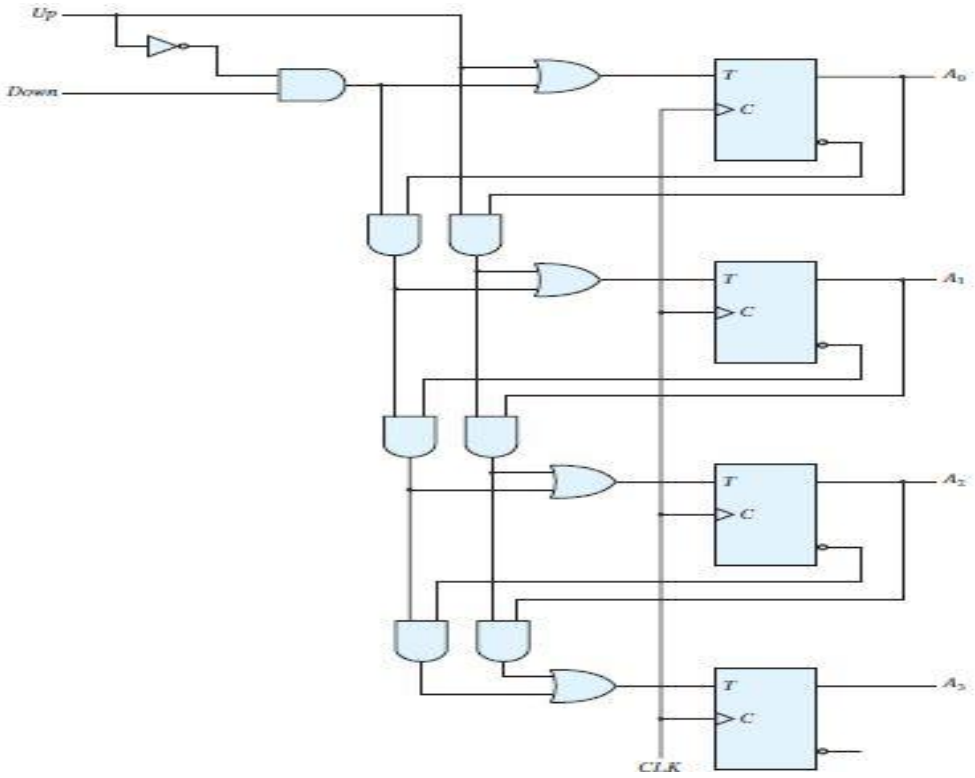



Figure 2: Up-Down Counter

3. Determine the operation of the circuit shown in Figure 3. Choose n = 3 and write down the sequence of outputs for 6 cycles. Now let n = 8. What is the (new) sequence ? Proceed until the sequence repeats.

4. What are the successive values of $Q_0$ and $Q_1$ in the circuit shown in Figure 4 ? Assume initial values for $Q_1$ and $Q_0$ to be 0 and 0 respectively.

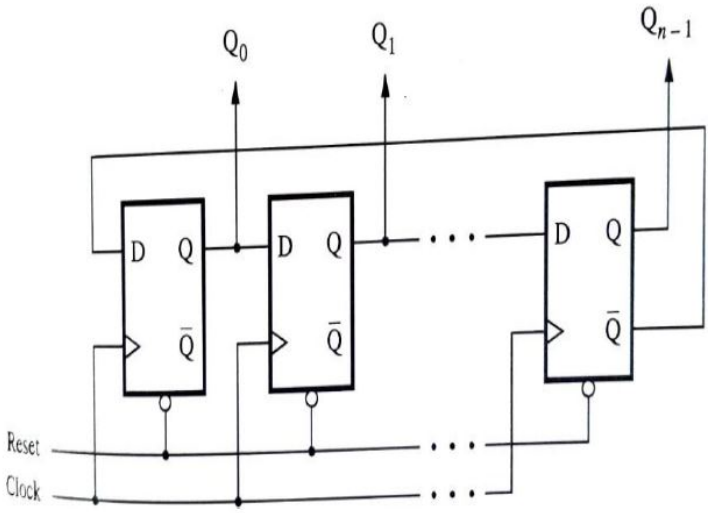



Figure 3: Another counter

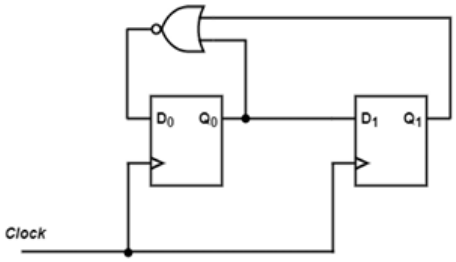



Figure 4: 2-bit Sequence

5. The circuit in Figure 5 operates like a counter. What is the counting sequence for this circuit ? Assume the initial values as follows: $Q_0 = Q_1 = Q_2 = 0$.

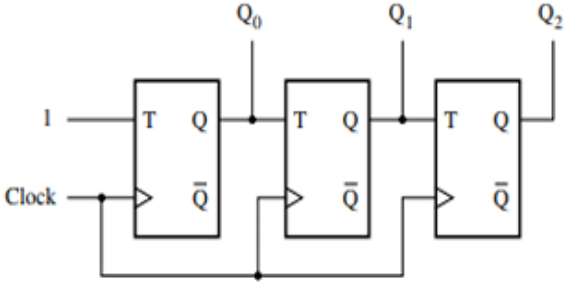



Figure 5: Three T Flip-Flops

6. Consider the sequential circuit shown in Figure 6. Find the maximum clock frequency (in MHz), at which the given circuit can operate reliably. Note that $t_{pd}$, $t_{setup}$ and $t_{hold}$ denote the propagation delay, setup time and hold time respectively.

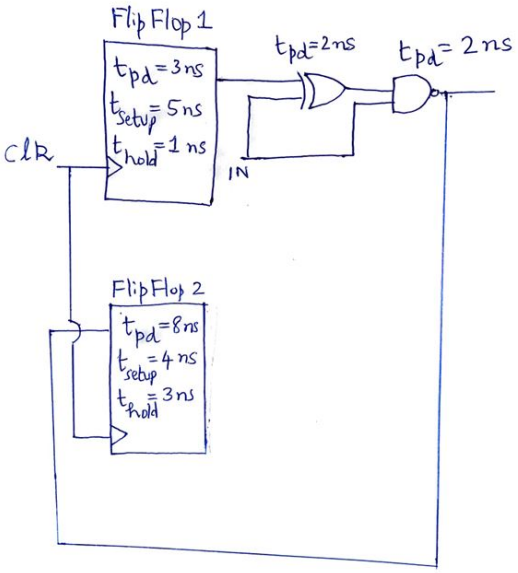



Figure 6: Circuit with specific values for propagation delay, setup time and hold time

7. Consider the synchronous sequential circuit in Figure 7. Analyse the circuit by writing the input equations. Also, draw the state transition table.

| | $Q_1$ | $Q_0$ |
|---|---|---|
| **A** : | 0 | 0 |
| **B** : | 0 | 1 |
| **C** : | 1 | 1 |
| **D** : | 1 | 0 |

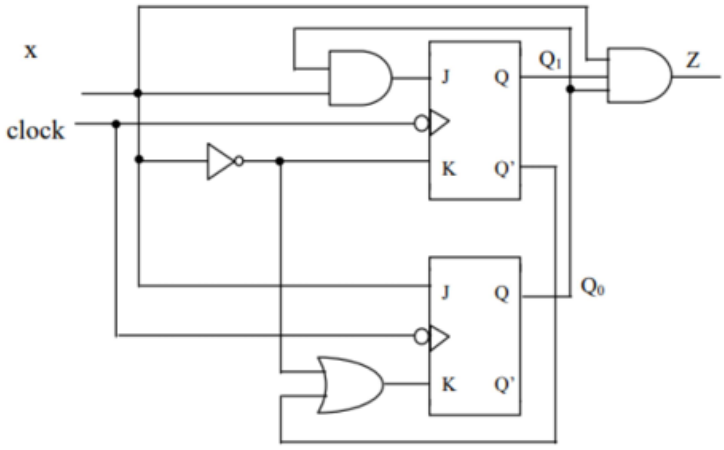
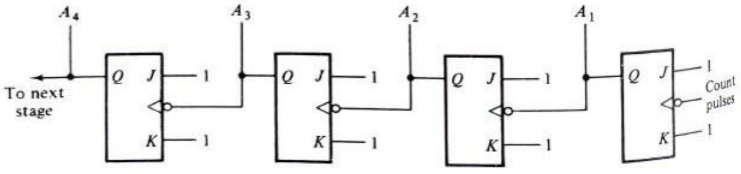
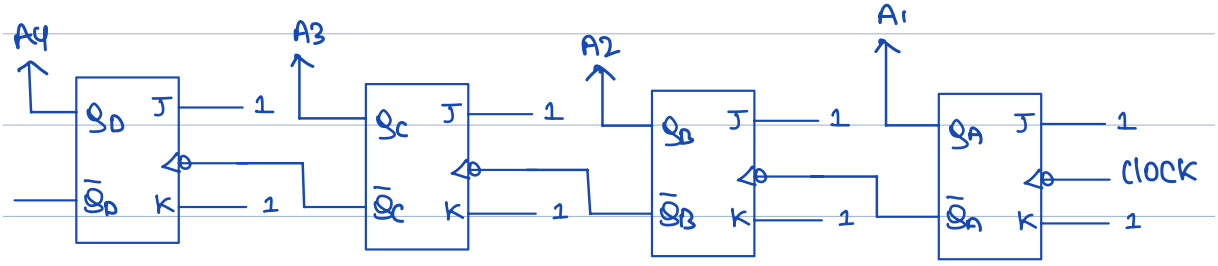



Figure 7: Synchronous Sequential Circuit

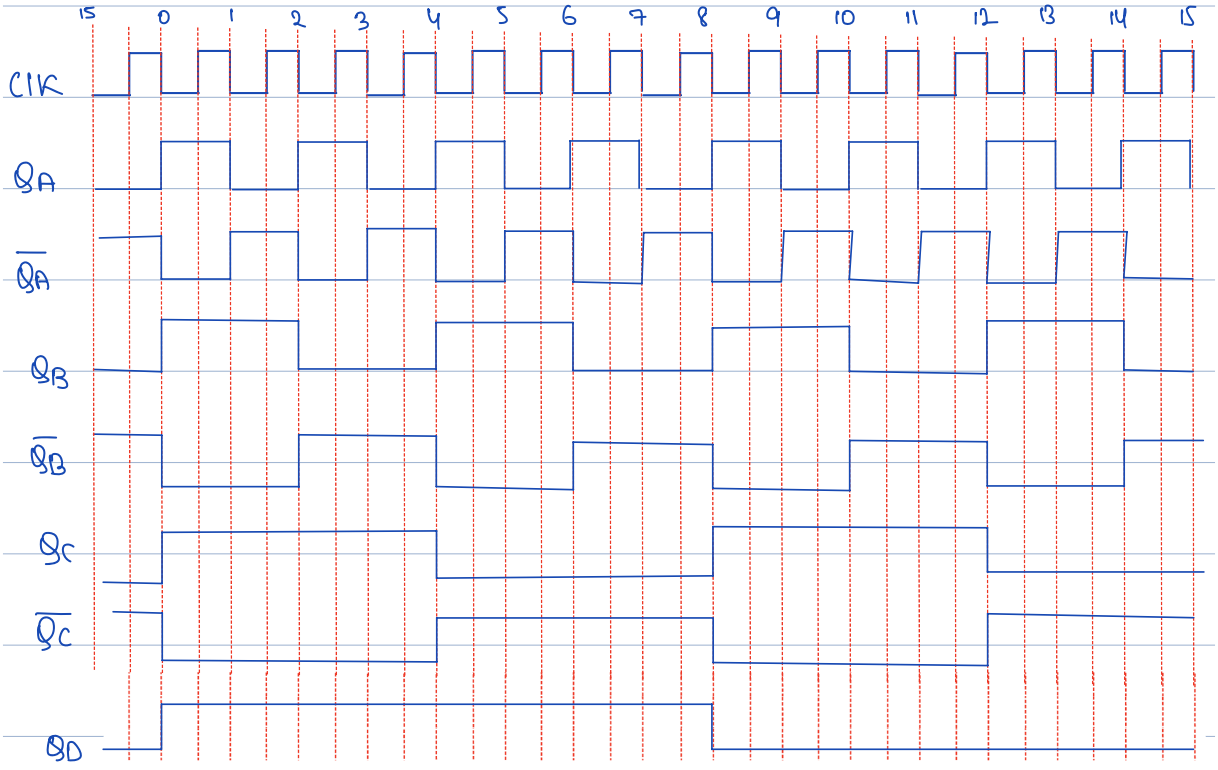
8. A T flip-flop has a 15ns delay from the time its clock (pulse) input goes from 1 to 0 to the time the output is complemented. What is the maximum delay in a 8-bit binary ripple counter that uses this type of flip flop? What is the maximum frequency at which the counter can operate reliably ?

1. Consider the circuit shown in Figure 1. We want this circuit to count down from 15 to 0. Suggest what changes are required and draw the new circuit. Also, explain the operation of your (new) circuit.



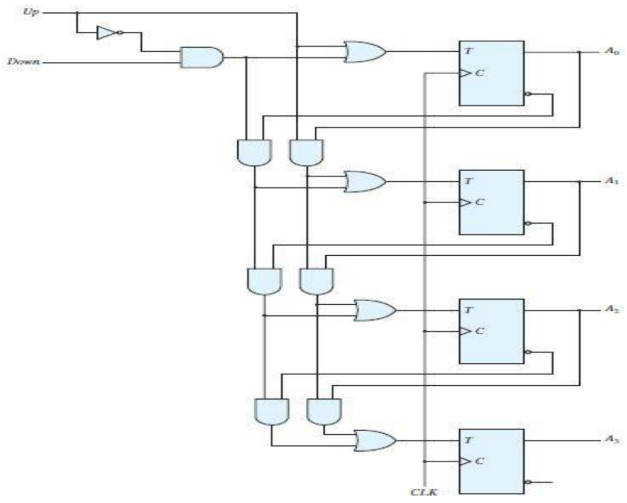




we can explain using timing diagram




If we check outputs $Q_A$ (LSB) to $Q_D$ (MSB) we see that it's a down counter from 15 (1111) to 0 (0000)

2. An Up-Down counter circuit is shown in Figure 2. Study the behaviour of this circuit for all possible inputs (binary values of Up and Down).




(1) UP=0 DOWN=0: The circuit doesn't change it's state

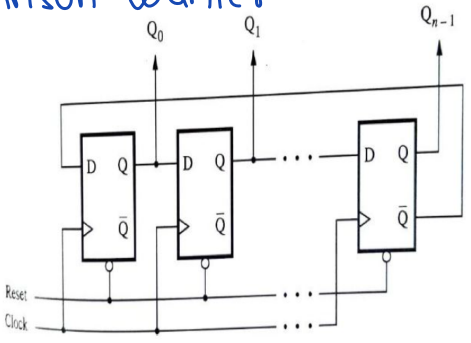
(2) UP=0 DOWN=1: Acts as a 4-bit down counter

(3) UP=1 DOWN=0: Acts as a 4-bit up counter

(4) UP=1 DOWN=1: Acts as a 4-bit up counter

3. Determine the operation of the circuit shown in Figure 3. Choose n = 3 and write down the sequence of outputs for 6 cycles. Now let n = 8. What is the (new) sequence ? Proceed until the sequence repeats.

The beside figure is a Johnson Counter



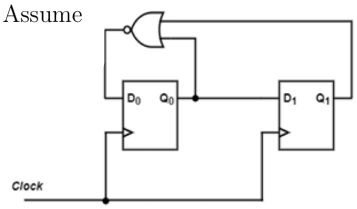

If n=3:

| t | $Q_0$ | $Q_1$ | $Q_2$ |
|---|---|---|---|
| | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 |
| 2 | 1 | 1 | 0 |
| 3 | 1 | 1 | 1 |
| 4 | 0 | 1 | 1 |
| 5 | 0 | 0 | 1 |
| 6 | 0 | 0 | 0 |

If n=8:

| t | $Q_0$ | $Q_1$ | $Q_2$ | $Q_3$ | $Q_4$ | $Q_5$ | $Q_6$ | $Q_7$ |
|---|---|---|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 5 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 |
| 7 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 8 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 9 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| 11 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 12 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 13 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 14 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 15 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 16 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

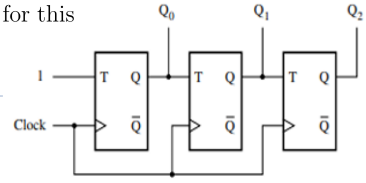
4. What are the successive values of $Q_0$ and $Q_1$ in the circuit shown in Figure 4 ? Assume initial values for $Q_1$ and $Q_0$ to be 0 and 0 respectively.




| t | $Q_0$ | $Q_1$ |
|---|---|---|
| | 0 | 0 |
| 1 | 1 | 0 |
| 2 | 0 | 1 |
| 3 | 0 | 0 |

Repeats at time interval 3

5. The circuit in Figure 5 operates like a counter. What is the counting sequence for this circuit ? Assume the initial values as follows: $Q_0 = Q_1 = Q_2 = 0$.

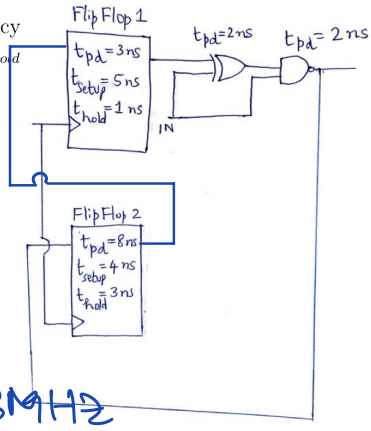



| t | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|
| | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 0 | 1 | 0 |
| 3 | 0 | 0 | 1 |
| 4 | 0 | 0 | 0 |

$Q_0$ toggles alternatively
$Q_1$ toggles when $Q_0$ is 1
$Q_2$ toggles when $Q_1$ is 1

Repeats after 4 intervals

6. Consider the sequential circuit shown in Figure 6. Find the maximum clock frequency (in MHz), at which the given circuit can operate reliably. Note that $t_{pd}$, $t_{setup}$ and $t_{hold}$ denote the propagation delay, setup time and hold time respectively.




3+2+2+4 = 11 ns for 1 to 2.

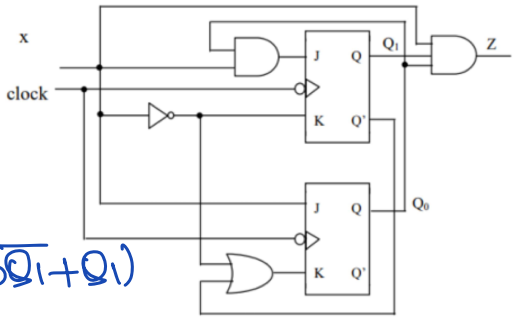
8+5 = 13 ns for 2 to 1

$\Rightarrow T = 13 \Rightarrow f = \frac{1000}{13}$ MHz

$= 76.923$ MHz

7. Consider the synchronous sequential circuit in Figure 7. Analyse the circuit by writing the input equations. Also, draw the state transition table.

| | $Q_1$ | $Q_0$ |
|---|---|---|
| **A** : | 0 | 0 |
| **B** : | 0 | 1 |
| **C** : | 1 | 1 |
| **D** : | 1 | 0 |




$J_1 = x Q_0 \quad K_1 = \overline{x}$

$J_0 = x \quad K_0 = \overline{x} + \overline{Q_1}$

$Q_1(t+1) = x Q_0 \overline{Q_1} + x Q_1 = x(Q_0 \overline{Q_1} + Q_1)$

$Q_0(t+1) = x \overline{Q_0} + x Q_1 (Q_0)$

$= x(Q_0 Q_1 + \overline{Q_0})$

| J | K | Q(t+1) |
|---|---|---|
| 0 | 0 | Q(t) |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | $\overline{Q(t)}$ |

characteristic equation of J-K flip flop

| present state | | input | Next State | | Output |
|---|---|---|---|---|---|
| $Q_1$ | $Q_0$ | $x$ | $Q_1^*$ | $Q_0^*$ | $z = x Q_0^* Q_1^*$ |
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 0 |

state transition table

8. A T flip-flop has a 15ns delay from the time its clock (pulse) input goes from 1 to 0 to the time the output is complemented. What is the maximum delay in a 8-bit binary ripple counter that uses this type of flip flop? What is the maximum frequency at which the counter can operate reliably ?

$t_{pd} = 15\,ns$

since the clock of other flip flops is the output of the previous flip flops, the delay driven by

adds up for each flip flop

Total delay = 15ns x 8 = 120ns

Best counter frequency = $\frac{1}{120ns} = \frac{1000}{120}$ MHz

$= 8.33$ MHz